

INVISIGUARD - INTELLIGENT SYSTEM FOR PREDICTING CYBER THREATS REAL TIME



Zero-Day Attack Prediction by Analyzing Network Traffic Patterns

Specialization: Cyber Security



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RP-25-26J-062

Research Problem

- Traditional security systems struggle to detect Zero-Day attacks because they rely on predefined signatures.
- Not only that, but current anomaly-based systems also often fail due to:
 - High volume of false positives.
 - Inability to adapt to new attack patterns.
 - Limited use of machine learning to detect unknown threats.

Research Gap

Application References	Zero-Day Prediction
Research 1	✓
Research 2	✓
Research 3	✗
InvisiGuard	✓



Key Domains

Threat Detection : How Zero day attacks work, understanding about IPS and SOC operations.

Traffic Analysis : Understanding about packet level features and flow level features

MITRE ATT&CK Framework : Mapping detected anomalies to real-world attacker tactics & techniques.

TOOLS AND TECHNOLOGIES

Tools:

- Jupyter Notebook - IDE & experimentation
- Git/GitHub - version control

Technologies:

- Python
- Deep Learning Frameworks – TensorFlow
- Visualization Dashboard – Streamlit or Dash (for web UI)
- Scikit-learn - Performance Evaluation



Solution Evaluation

Measure Metrics

- Accuracy → correct predictions
- Precision → flagged anomalies that are real
- Recall → attacks detected
- F1-Score → balance of precision and recall
- False Positive Rate → normal traffic wrongly flagged

Overall Percentage

Output lower false positives
with 98% detection rate

Data Availability & Ethical Clearance

Sources

Publicly available anomaly
detection datasets.

- HIKARI 2021 in Zenodo
- UNSW-NB15 V3 in Zenodo

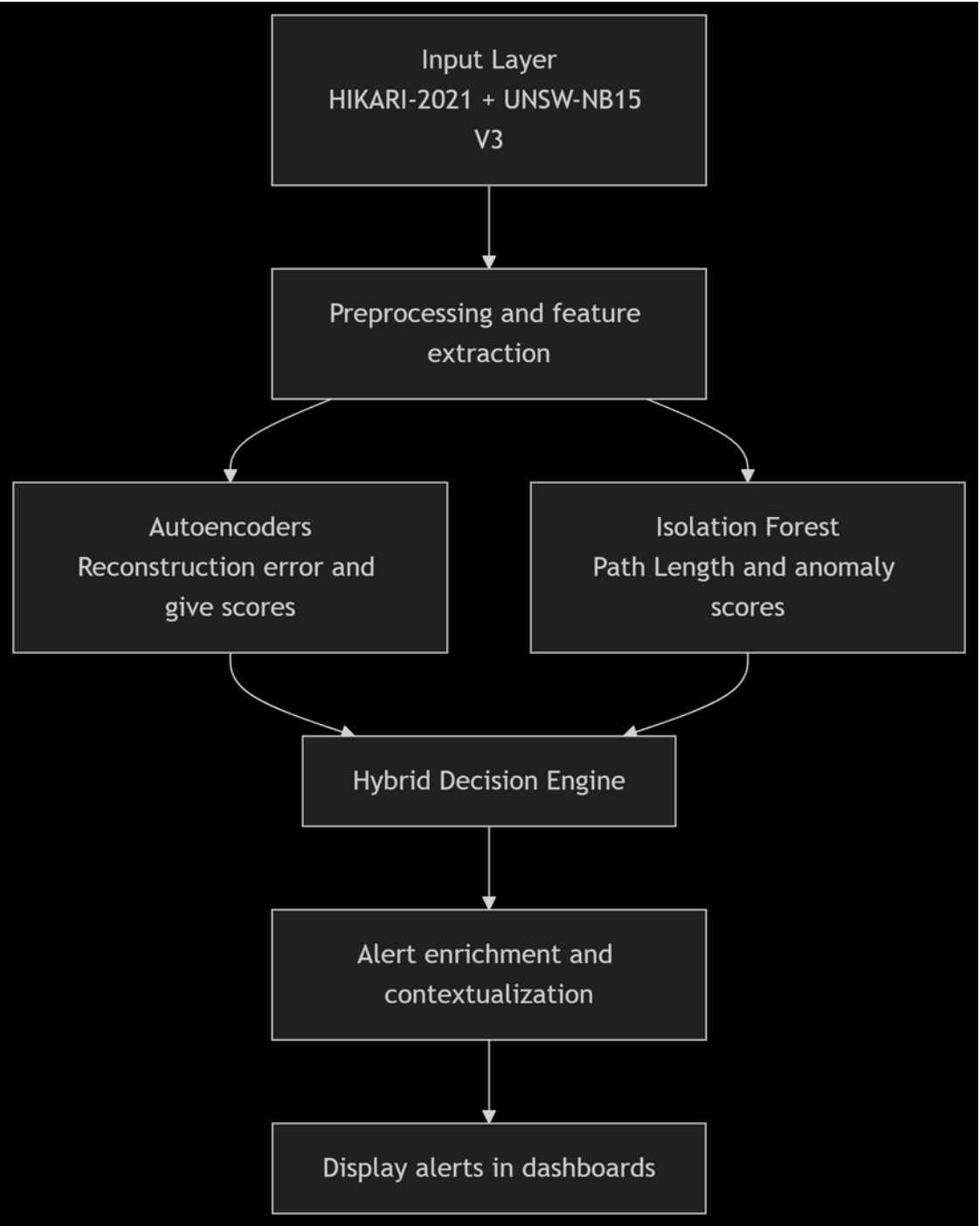
Ethical & Legal Compliance

All research data has been
anonymized, and no sensitive or
private information is used here.



Zero-Day Attack Prediction by Analyzing Network Traffic Patterns

Component Diagram



Real-World Use Cases

Banking Sector

- Autoencoder spots irregular transaction timing → Isolation Forest isolates outlier user activity
- Risk Score integrates into access policies, revoking privileges in real-time

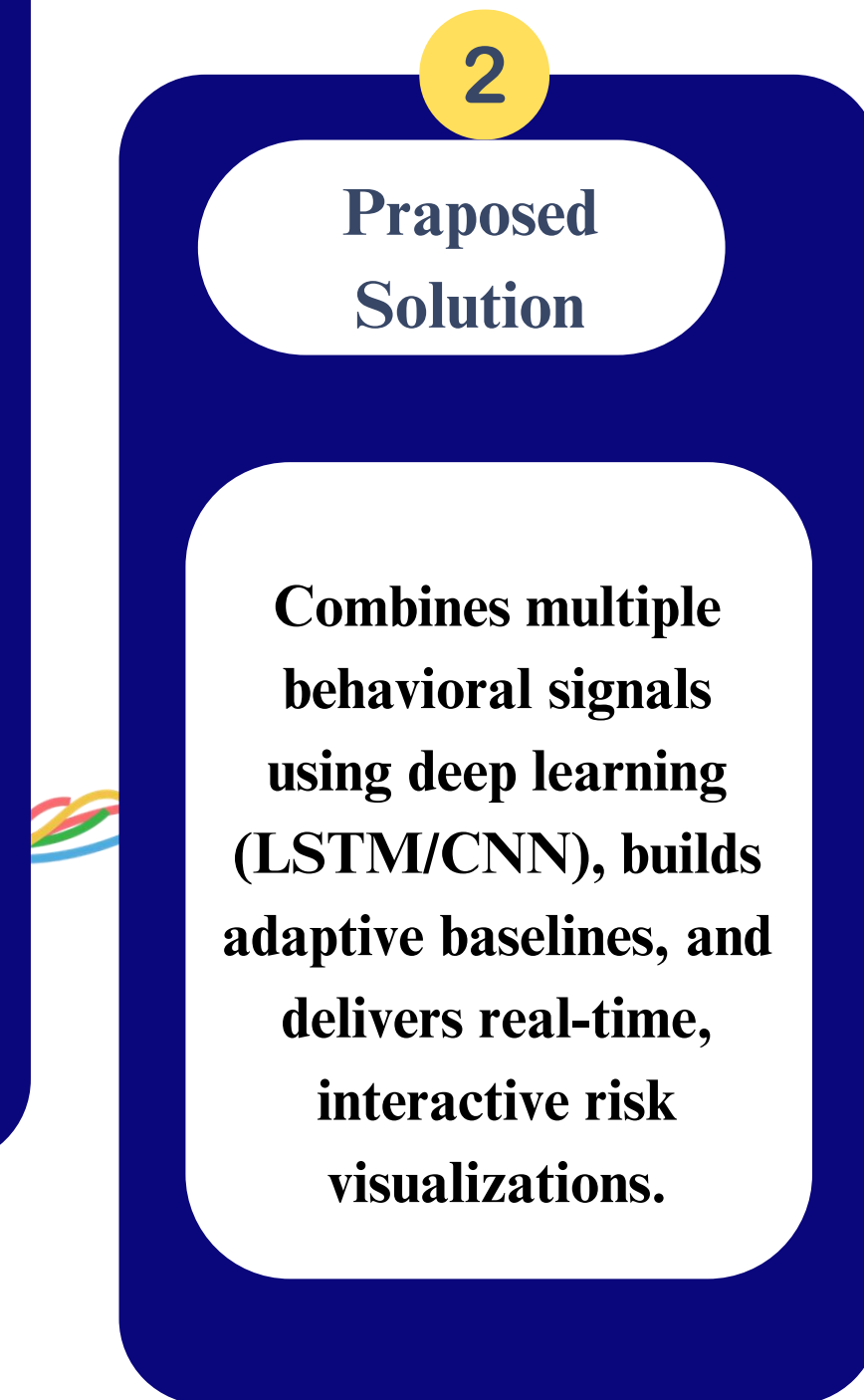
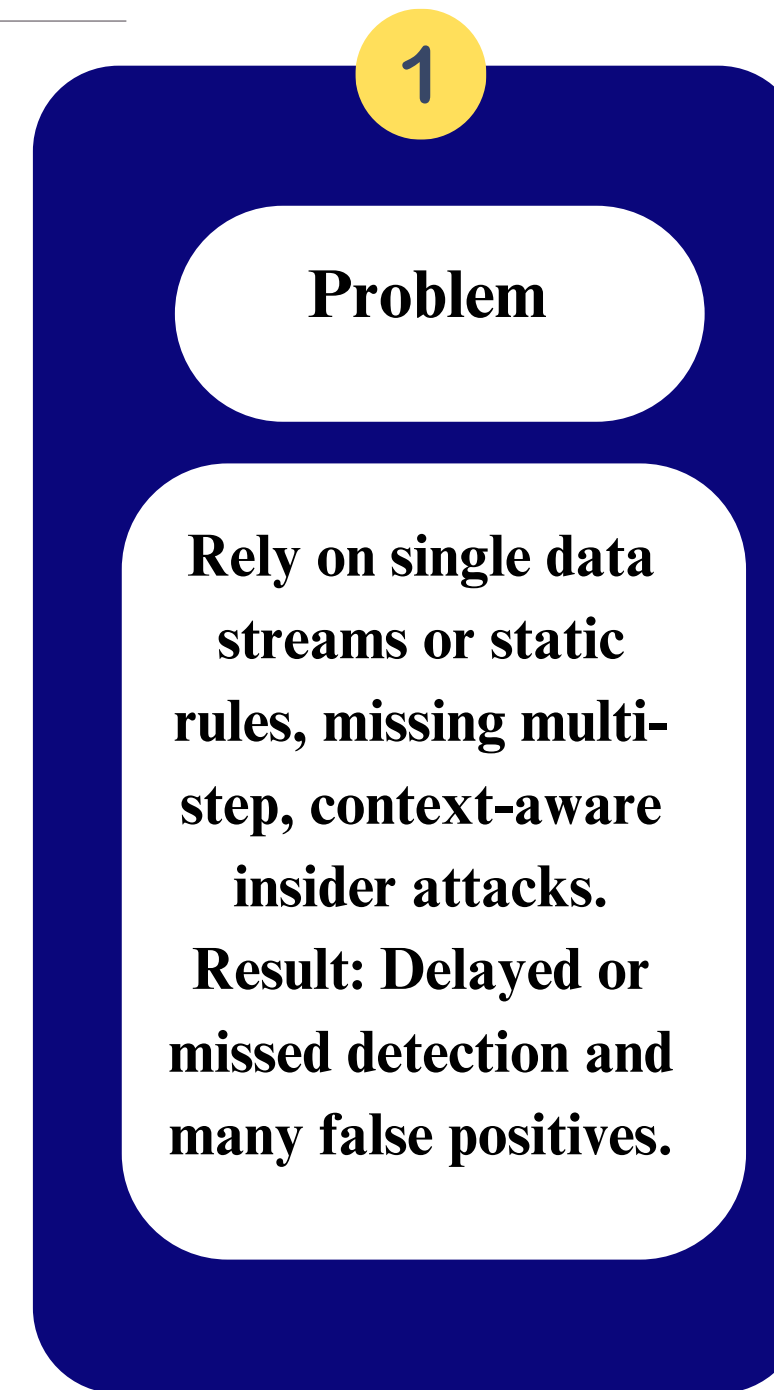
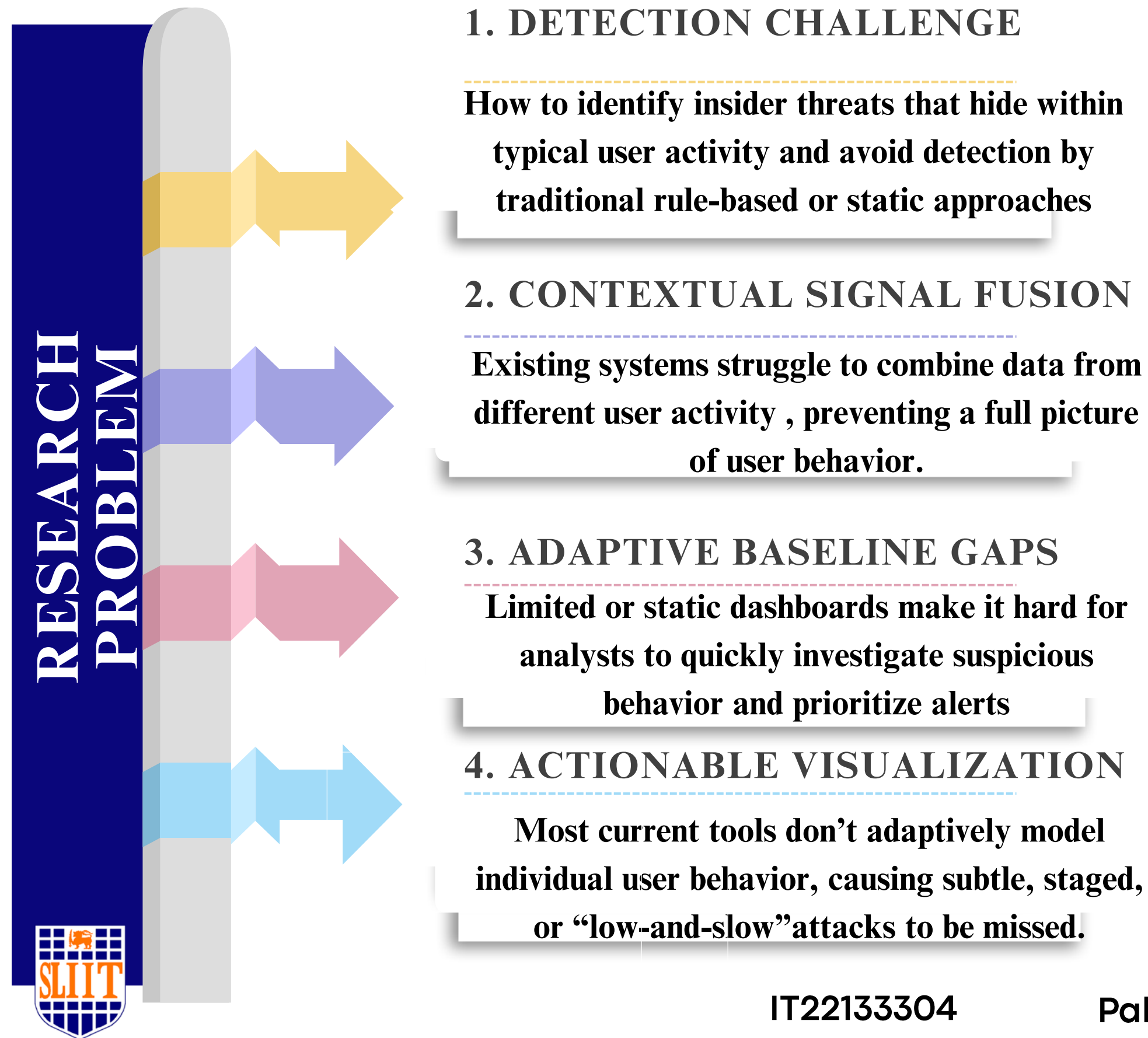
SOC Operations

- Captures live traffic and generates real-time alerts for potential Zero-Day attacks, while providing information mapped to MITRE ATT&CK techniques and severity levels for better understanding by SOC analysts.

Predicting Known Insider Attacks Using User Behavioral Patterns

Information Technology(Specialization in Cyber Security)





PREDICTING KNOWN INSIDER ATTACKS USING USER BEHAVIOURAL PATTERNS

RESEARCH GAP

Feature / Capability	Splunk UBA	Proofpoint ITM (ObserveIT)	Forcepoint Insider Threat	
Behavioral Signal Fusion	✗	✗	✗	
Deep Learning (LSTM/CNN) for Patterns	✗	✗	✗	
Personalized Baseline Modeling	✗	✗	✗	



Key Domains of Specialization Knowledge

Cyber Security

Strengthens enterprise security by detecting abnormal user activities, insider misuse, and risky behavior using behavioral analytics.

Machine Learning

Employs advanced deep learning models (LSTM, CNN) to learn complex, sequential user activity and accurately spot patterns typical of insider threats.

Risk Analysis

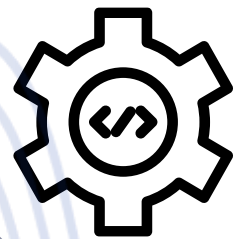
Translates detected anomalies and model outputs into actionable risk scores and intelligence for security teams.

Access Control

Integrates alert data and risk scores with organizational policies to enable dynamic adjustment of user privileges and access in real time.

Visualization

Provides interactive dashboards (Streamlit/Dash) for monitoring behavioral trends, alerts, and supporting forensic investigations.



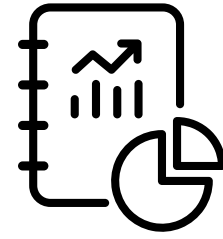
Data Privacy and Anonymization

All research data is fully anonymized; no personal or sensitive information is included.



Data Sources

Utilizes open-source and publicly available datasets



Compliance

Compliance with GDPR & ISO 27001 for any real access logs.



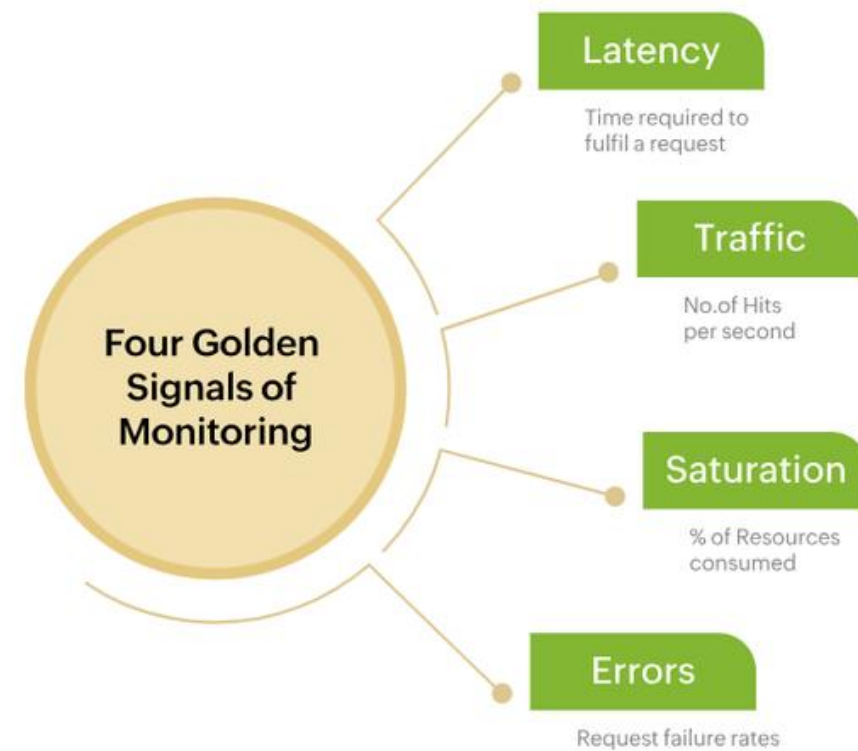
Data Availability and Ethical Clearance

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Evaluation and Validation Metrics for Measuring Success



Real World Analysis

01

Data Sources

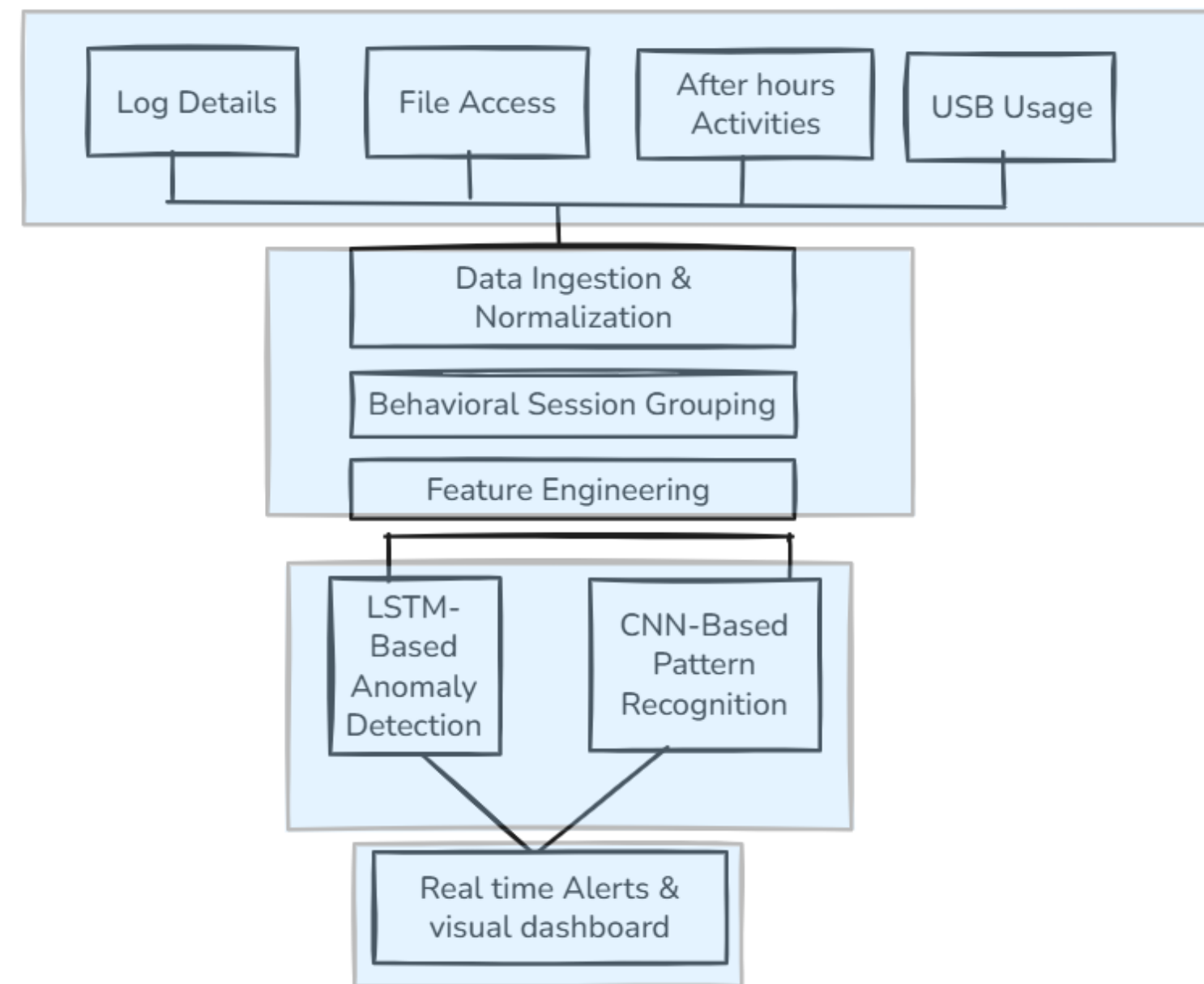
- Intrusion Detection Dataset
<https://www.kaggle.com/dnkumars/cyber-security-intrusion?detectiondataset>
- Insider threat detection -
<https://www.kaggle.com/mrajaxnp/cert?insiderthreat-detection-research>

02

Latest Technologies

- TensorFlow Keras
- NetworkX
- Visual Studio Code
- AWS / Google Cloud Platform

Component



Real World Scenario

Continuous Data Collection

The system automatically gathers logs of user activities

Data Processing & Feature Extraction

Logs are cleaned, processed, and transformed into behavioral features

Establishing Behavioral Baselines

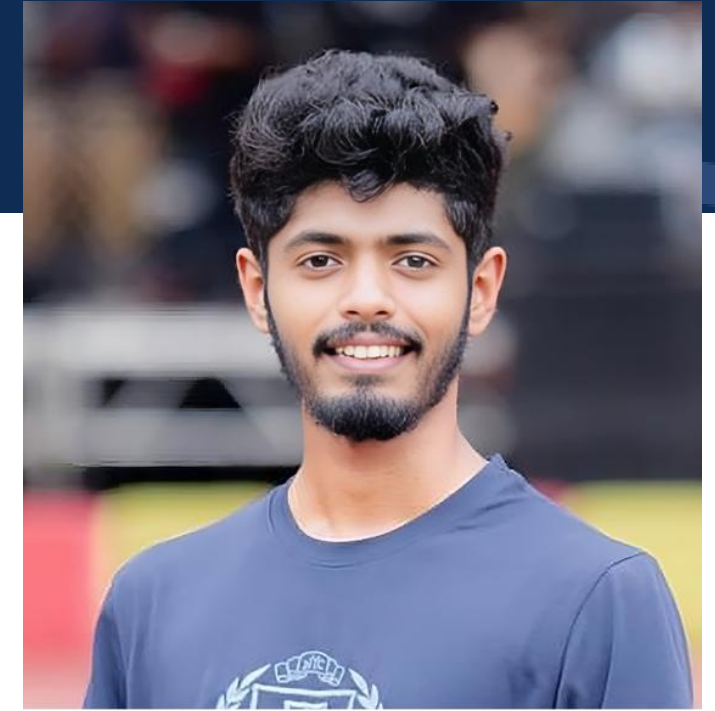
Over time, the system creates a personalized baseline

Pattern Recognition Using Deep Learning

The LSTM and CNN models analyze the sequence and pattern of these unusual activities

Anomaly Scoring and Alert Generation

The system assigns a risk score



KNOWN ATTACK PREDICTION BY ANALYZING ENDPOINT LOG PATTERNS

Information Technology(Specialization in Cyber Security)

KNOWN ATTACK PREDICTION BY ANALYZING ENDPOINT LOG PATTERNS

Research Problem

In today's dynamic cybersecurity environment, traditional signature-based and rule based intrusion detection systems struggle against sophisticated attacks and advanced persistent threats. The sheer volume of endpoint devices and log data creates an urgent need for intelligent, scalable security solutions.

The Research Gap: Why Current Solutions Fall Short

Reactive Dependence

Most systems rely on static signatures & predefined rules, creating a vulnerability window for new attack variants.

Insufficient Hybrid Architectures

Lack of integrated multi-stage solutions balancing speed and precision for detection.

Lack of Proactive Prediction

Systems identify attacks post-occurrence, missing subtle precursor patterns.

Scalability

Bottlenecks

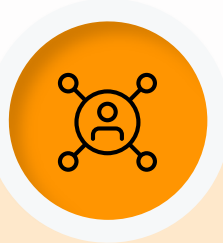
Massive log volumes challenge real-time analysis, leading to delays and computational expense.

Limited Log Exploitation

Limited exploitation of endpoint logs for predictive purposes.

KNOWN ATTACK PREDICTION BY ANALYZING ENDPOINT LOG PATTERNS

Proposed Solution



Unified Log Analysis

Utilize Windows and Linux system endpoint log data for real-time cyberattack identification.



NLP Feature Engineering

Employ TF-IDF to convert raw log data into meaningful representations, extracting detailed event information.

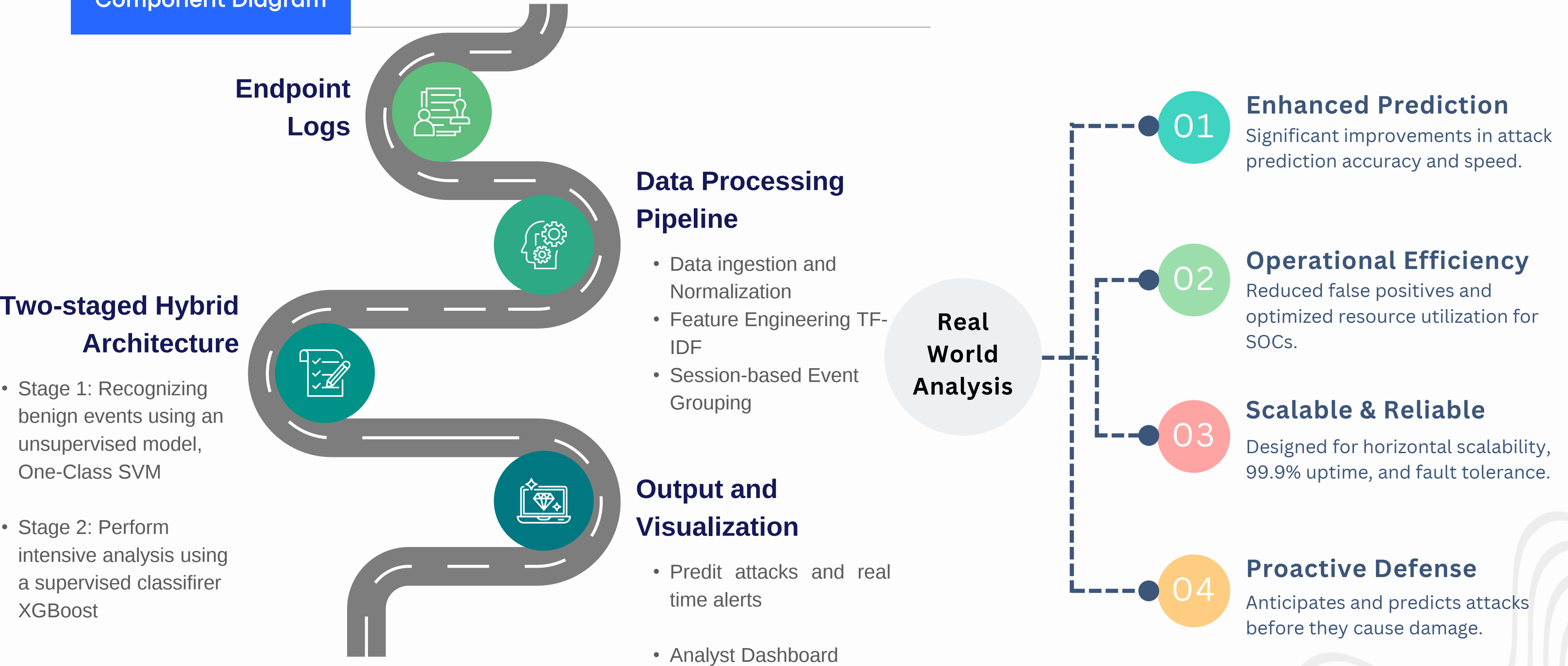


Two-Stage ML Architecture

Combine One-Class SVM for high-speed filtering with XGBoost for precise, supervised classification of known attacks.

Application Reference	Unstruected Text Inputs	Cr
Research 1	✓	
Research 2	✓	
Research 3	✓	
Proposed System	✓	





Key Domains of
Specialization

Cybersecurity & Threat Detection

Endpoint monitoring, intrusion detection, SOC operations.

Operating Systems

Parsing Event Logs (EVTX, Sysmon, Syslog, auditd)

Machine Learning & Data Science

NLP-based feature extraction, hybrid anomaly filtering + classification.

Log Analysis & Event Correlation

Session-based sequencing, cross-log contextualization.

Latest
Technologies

NLP Techniques

TF-IDF Vectorization of log sequences.

Hybrid ML Pipeline

One-Class SVM + XGBoost

Log Sources

Windows EVTX/Sysmon, Linux auditd & Syslog, registry/file integrity monitoring

Threat Framework

Mapping alerts to MITRE ATT&CK tactics & techniques.



References

- [1] K. Patel, S. Singh, and M. Bhatia, "A scalable hybrid framework for enterprise log analysis," in Proc. 10th Int. Conf. on Internet of Things, Big Data and Security (IoTBDs), Scitepress, 2025.
- [2] N. A. Nipa, N. Bouguila, and Z. Patterson, "A Comparative Study of Log-Based Anomaly Detection Methods in Real-World System Logs," in Proc. 10th Int. Conf. on Internet of Things, Big Data and Security (IoTBDs), 2025, pp. 141–152, doi: 10.5220/0013367000003944.
- [3] N. B. Babu and M. Gunasekaran, "An Analysis of Insider Attack Detection Using Machine Learning Algorithms," in 2022 IEEE 2nd Int. Conf. on Mobile Networks and Wireless Communications (ICMNWC), 2022, pp. 1–8, doi: 10.1109/ICMNWC56175.2022.10032009.



KNOWN ATTACK PREDICTION BY ANALYZING SYSTEM CALL PATTERNS



KNOWN ATTACK PREDICTION BY ANALYZING SYSTEM CALL PATTERNS

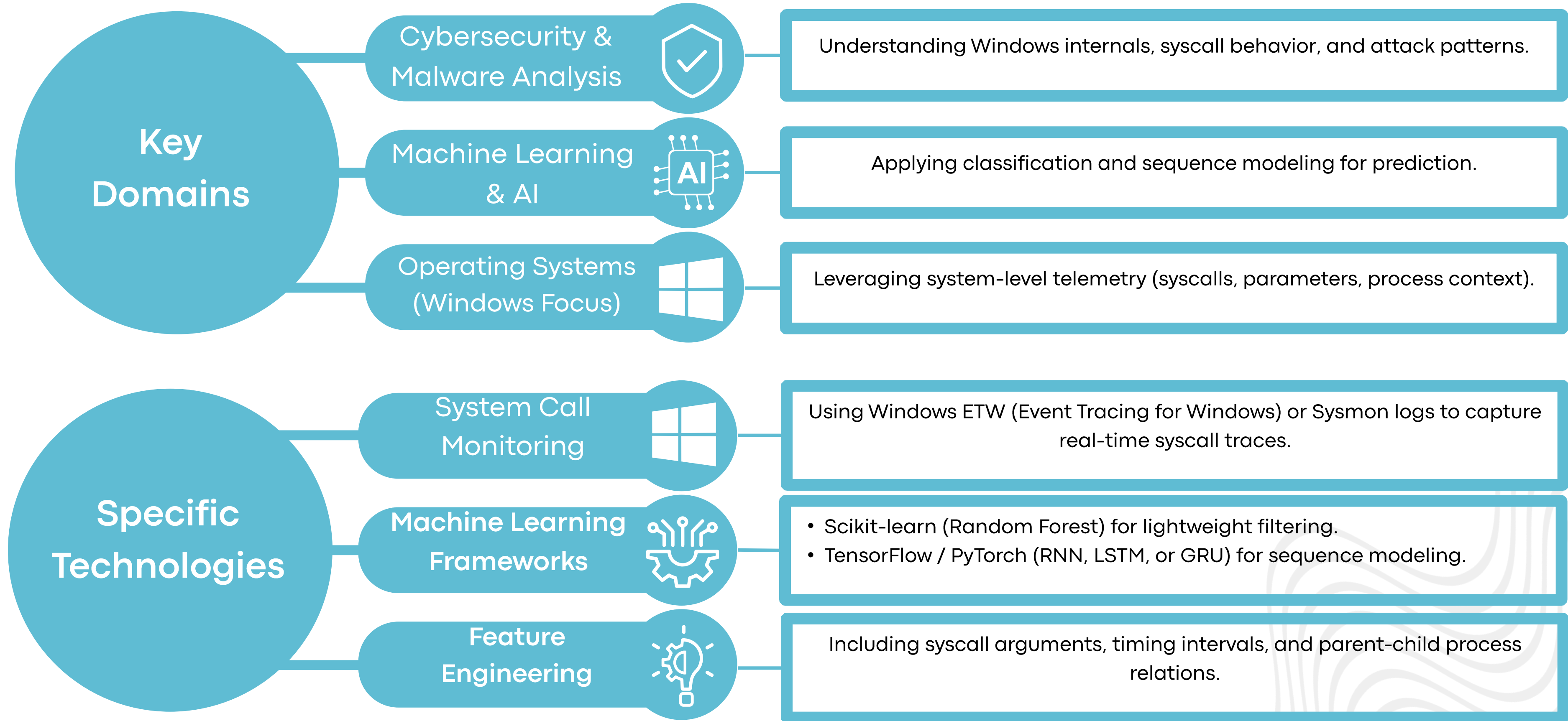
Problem - Current Windows malware detection is either black-box and heavyweight (EDRs) or lightweight but limited (academic syscall models), failing to provide early, context-rich, transparent prediction.

Solution - A lightweight, hybrid ML framework that predicts malicious intent from system-call patterns + context in real time, bridging the gap between practical deployability and research transparency.

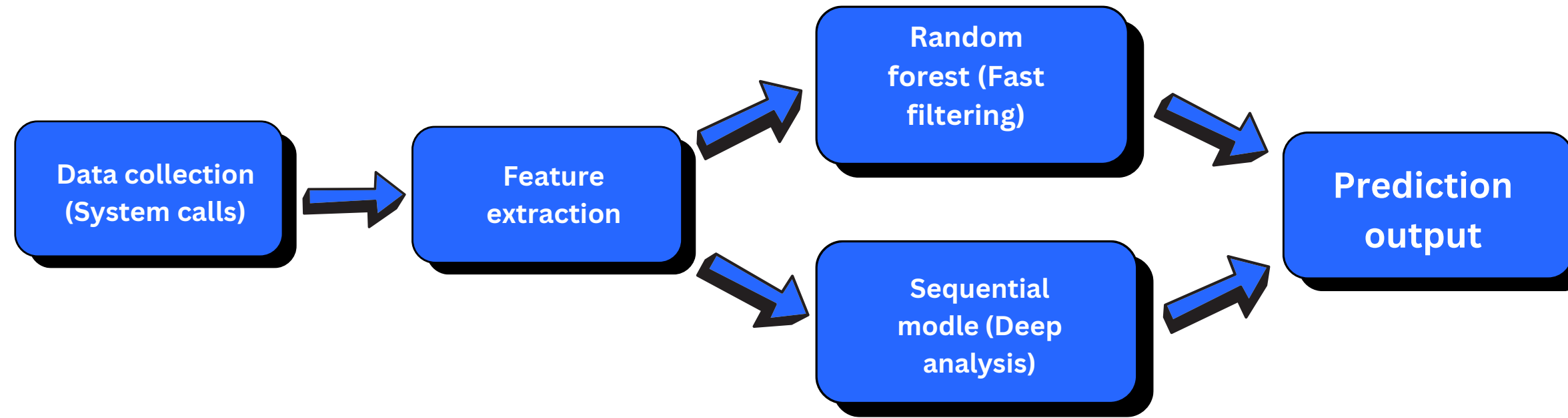
NtMalDetect	Systemcall SVM	InvisiGuard
- Limited context, no early prediction, simpler models - No parameter info, limited sequence modeling, no early prediction, separates supervised/unsupervised.	- Lacks long-range sequence modeling bag-of-n-grams captures short-term patterns but not complex temporal dependencies. - Not designed for early-stage prediction; it's focused on detection post-execution or mid-execution.	- Parameter-Rich Modeling - Deep Sequence Modeling - Early-Stage Prediction Capability - Hybrid Supervised + Unsupervised Detection - Explainability for Trust



KNOWN ATTACK PREDICTION BY ANALYZING SYSTEM CALL PATTERNS



KNOWN ATTACK PREDICTION BY ANALYZING SYSTEM CALL PATTERNS



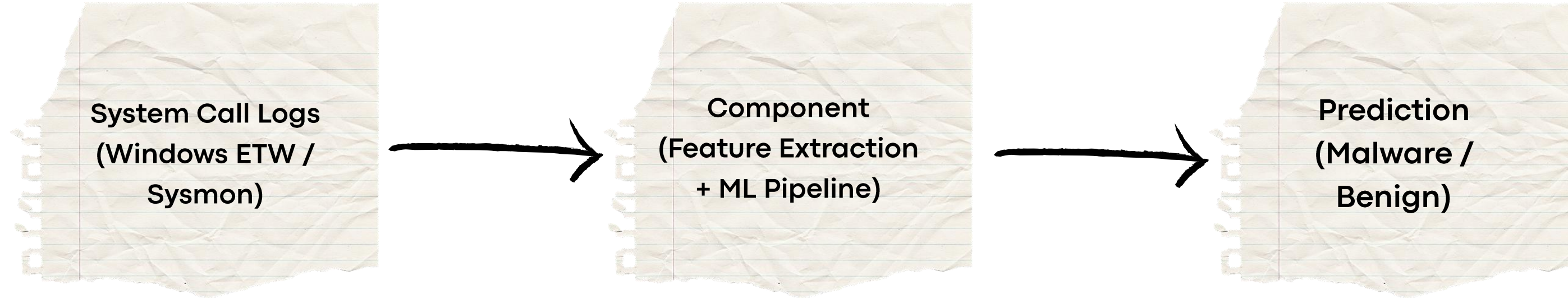
Data availability

- 1) Risk-Based System-Call Sequence Grouping Method for Malware Intrusion Detection
Link - <https://github.com/DjPasco/AWSCTD>
- 2) ADFA-LD dataset (system call traces for intrusion detection).
- 3) <https://data.mendeley.com/datasets/kbt8xt3678/3>

Ethical Clearance

- 1) Experiments run only in isolated virtual environments (VMs / sandboxes).
- 2) Malware samples sourced legally from academic repositories.
- 3) No harm to production systems, networks, or third parties.
- 4) Focus is strictly academic research and defense-oriented innovation.

KNOWN ATTACK PREDICTION BY ANALYZING SYSTEM CALL PATTERNS



How to Use in Real-World Scenario (right side / bottom half with icons):

- Endpoint Protection Agent → Installed on Windows laptops/desktops.
- Continuous Monitoring → Collects real-time system call logs.
- Prediction Engine → Detects malware before it executes fully.
- Action Taken → Alerts security team OR automatically blocks the process.

InvisiGuard: Commercialization

- Integrates with SIEM/SOAR; targets the \$25B+ Enterprise SOC and Threat Detection market.
- The global AI in cybersecurity market is projected to reach USD 100+ billion by 2030.0
- Offer as a modular SaaS cybersecurity solution to Medium to Large enterprises across Finance, Healthcare, Technology, and Government sectors.

Target Market

- Market: Entry level to mid Enterprises (Finance, Healthcare, Government).
- Customers: CISOs and SOC Managers in mid employee organizations.
- Value: Drastically reduces alert fatigue and MTTR with high-fidelity, correlated alerts.
- IPR: Patent on the hybrid correlation framework; copyright on proprietary ML models and code.

Cost

InvisiGuard: Commercialization

- GCP/Azure hosting, compute for ML models, and storage (~ LKR 2,000 – 5,000 / month).
- Commercial threat intelligence feeds, dataset access (~ LKR 1,000 annually).
- Continuous updates, DevOps, support team (~ LKR 3,000 – 6,000 / month).

**Thank
You**